

## Atomistic Simulations of Nanoindentation Response of Irradiation Defects in Iron (Simulasi Atomistik bagi Tindak Balas Pelekukan Nano bagi Kecacatan Penyinaran dalam Besi)

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### ABSTRACT

*Radiation response of a material is a consequence of defects' evolution in any radiation damage event. The radiation-induced defects can significantly alter the mechanical properties of a material. Radiation damage initiates from incident neutron by bombardment on solid material causing production and evolution of Frenkel defects. Since voids are formed due to aggregation of a large number of vacancies that cause dimensional changes and hence irradiation-induced swelling. In order to characterize the effect of irradiation defects, we have performed molecular dynamics (MD) simulations to investigate nanoindentation response of point defects and voids in Fe and their effects on mechanical parameters. The radial effect of voids and their interaction mechanism is also explored by nanoindentation simulation. It has been found that most of the dislocation produced are  $\frac{1}{2} \langle 111 \rangle$  and  $\langle 100 \rangle$  during nanoindentation in all simulated models. There will be an increase in dislocation density which will harden the material and reduce its toughness. The mechanical parameters such as hardness  $H$  and reduced elastic modulus  $E_r$  of irradiation defects are calculated from P-h curves. It is found that both  $H$  &  $E_r$  of the point defects and voids are lower than the perfect model.*

*Keywords: Elastic modulus; hardness; irradiation defects; molecular dynamics simulations; nanoindentation*

### ABSTRAK

*Tindak balas sinaran sesuatu bahan adalah kesan daripada evolusi kecacatan dalam kejadian kerosakan sinaran. Punca kecacatan aruhan sinaran dengan ketara boleh mengubah sifat mekanik bahan. Kecacatan sinaran yang dimulakan daripada kejadian neutron dengan pembedilan pada bahan pepejal menyebabkan pengeluaran dan evolusi kecacatan Frenkel. Lompang terbentuk kerana pengagregatan sebilangan besar kekosongan yang menyebabkan perubahan dimensi dan bengkak teraruh penyinaran. Untuk mencirikan kesan kecacatan penyinaran, simulasi molekul dinamik (MD) telah dijalankan untuk mengkaji tindak balas pelekukan nano kecacatan titik dan lompang pada Fe dan kesannya terhadap parameter mekanik. Kesan jejari lompang dan mekanisme interaksinya juga diterokai oleh simulasi pelekukan nano. Didapati bahawa kebanyakan kehelan yang dihasilkan adalah  $\frac{1}{2} \langle 111 \rangle$  dan  $\langle 100 \rangle$  semasa pelekukan-nano dalam semua model simulasi. Terdapat peningkatan dalam ketumpatan perkehelan yang akan mengeraskan bahan dan mengurangkan keliatan. Parameter mekanik seperti kekerasan  $H$  dan pengurangan sinaran kecacatan modulus elastik  $E_r$  akan dihitung bermula dari lengkung P-h. Didapati bahawa kedua-dua titik cacat dan lompang  $H$  &  $E_r$  adalah lebih rendah berbanding model yang sempurna.*

*Kata kunci: Kecacatan sinaran; kekerasan; modulus elastik; pelekukan nano; simulasi molekul dinamik*

### INTRODUCTION

Radiation damage (RD) is a critical and challenging phenomena to be catered in structural materials of advanced reactors (Azevedo 2011). Radiation-induced microstructural evolution leads to swelling, irradiation growth and hence, changes in mechanical properties (Zinkle & Was 2013). In nuclear power industry, Fe-based alloys are extensively used as structural and cladding materials due to their excellent mechanical strength (Tan et al. 2016; Yvon & Carré 2009). Ferritic alloys have excellent void swelling resistance under ion irradiation (Yabuuchi et al. 2013). Long-term irradiation and high temperatures may cause extended defects like voids in the interior of the material that can change microstructure. Irradiation swelling depends on dislocation density and size of voids and precipitates (Renault et al. 2015). Voids and dislocation

loops are sources of void swelling and dimensional instabilities in the material (Bacon et al. 1994). Voids and bubbles are in the form of 3D vacancy clusters. Voids are vacancies which are independent of internal stresses while cavities are pressurized vacancies.

Nanoindentation also known as depth-sensitive technique (Ruestes 2017). It is one of the simplest and perhaps most common method for testing the mechanical properties of materials (Christopher et al. 2001). Prismatic dislocation loops are formed in metals after nanoindentation as seen by atomistic studies in BCC metals (Dalmau et al. 2012). Atomistic simulations can depict detailed insights into the physical and chemical phenomena to validate experimental results at the nanoscale (Uberuaga et al. 2018).

Nanoindentation performed by Richter et al. (2008) showed that hardness was increased in the presence of